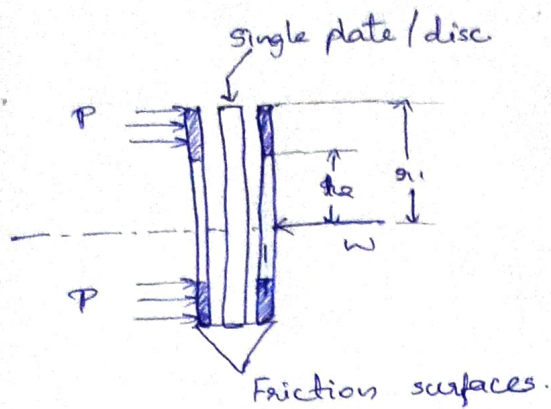


Design of single plate clutch.



w = Axial thrust.

P = Max intensity of pressure.

r_1 = Outer radi

r_2 = Inner radi

T = Torque transmitted.

P = Power.

N = Rotational rpm.

$$1. \quad T = \frac{P \times 60}{2\pi N}$$

$$2. \quad R_m = \frac{2}{3} \left[\frac{r_1^3 - r_2^3}{r_1^2 - r_2^2} \right]$$

$$3. \quad T = \mu w R_m$$

$$4. \quad P_{max} @ \text{outer radi, } r_2 \neq w = 2\pi C (r_1 - r_2)$$

1. A single plate clutch, effective on both side is required to transmit 25 kW @ 3000 rpm. Determine the outer & inner radii of frictional surface. The coefficient of friction is 0.255 & ratio of dia is 1.25 & the max pressure is not to exceed 0.1 N/mm^2 . Also determine the axial thrust to be provided by the spring, assuming theory of uniform wear.

Ans. Power, $P = 25 \text{ kW}$, $N = 3000 \text{ rpm}$, $\mu = 0.255$

$$\frac{d_1}{d_2} = 1.25 \quad \therefore P_{max} = 0.1 \text{ N/mm}^2$$

$$T = \frac{P \times 60}{2\pi N} = \frac{25 \times 10^3 \times 60}{2\pi \times 3000} = \underline{79.6 \text{ Nm}}$$

$$P_{max} r_2 = C \Rightarrow 0.1 r_2 = C$$

$$w = 2\pi C (r_1 - r_2) \\ = 2\pi (0.1 r_2) (r_1 - r_2)$$

$$\begin{aligned}
 W &= 2\pi (0.1 r_2) (1.25 r_2 - r_2) \\
 &= 2\pi \times 0.1 r_2 \times 0.25 r_2 \\
 &= 6.28 \times 0.025 r_2^2 \\
 &= 0.157 r_2^2
 \end{aligned}$$

$$\begin{aligned}
 T &= \eta \mu W R \\
 79.6 &= 2 \times 0.25 \times 0.157 r_2^2 \times \left(\frac{1.25 r_2 + r_2}{2} \right) \\
 &= 0.071 r_2^2 \cdot \frac{2.25 r_2}{2} \\
 &= 0.071 r_2^2 \cdot 1.125 r_2
 \end{aligned}$$

$$79.6 = 0.079 r_2^3$$

$$r_2^3 = \frac{79.6}{0.079}$$

$$\therefore r_2 = 96 \text{ mm}$$

$$\therefore r_1 = 120 \text{ mm}$$

$$W = 0.157 r_2^2 = 0.157 \times 96^2 = 1446.9 \approx 1447 \text{ Nmm}$$

2. A dry single plate clutch is to be designed for an automotive vehicle whose engine is rated to give 100 kW at 2400 rpm & the max torque 500 Nm. The outer radius of friction plate is 25% more than inner radii. The intensity of pressure p on the plate piece is not to exceed 0.07 N/mm^2 & may be assumed as 0.3 . The helical spring required by this clutch to provide axial force necessary to engage the clutch are 8. If each spring has a stiffness equal to 40 N/mm , determine the dimensions of friction plate & initial compression in the spring.

Ans. $P = 100 \text{ kW}$, $N = 2400 \text{ rpm}$, $T_{\text{max}} = 500 \text{ Nm}$,
 $r_1 = 1.25 r_2$, $p_{\text{max}} = 0.07 \text{ N/mm}^2$, $\mu = 0.3$
 No. of helical spring = 8, $R = \frac{r_1 + r_2}{2}$
 Spring stiffness, $k = 40 \text{ N/mm}$.

$$T = \frac{P \cdot 60}{2\pi N} = \frac{100 \times 10^3 \times 60}{2\pi \cdot 2400} = 397.8 \approx 398$$

$$\begin{aligned}
 C &= p_{\text{max}} r_2 \\
 &= 0.07 r_2
 \end{aligned}$$

$$\begin{aligned}
 W &= 2\pi C (r_1 - r_2) \\
 &= 2\pi \cdot 0.07 r_2 (r_1 - r_2) \\
 &= 2\pi \cdot 0.07 r_2 (1.25 r_2 - r_2) \\
 &= 6.28 \times 0.07 r_2 \times 0.25 r_2 \\
 &= 0.109 r_2^2 \\
 W &= 0.11 r_2^2
 \end{aligned}$$

$$\begin{aligned}
 T &= \eta \mu W R \\
 &= 2 \times 0.3 \times 0.11 r_2^2 \times \left(\frac{1.25 r_2 + r_2}{2} \right) \\
 &= 0.6 \times 0.11 r_2^2 \times 1.125 r_2
 \end{aligned}$$

$$500 = 0.074 r_2^3$$

$$r_2 = 190 \text{ mm}, \quad r_1 = 237 \text{ mm}$$

$$\begin{aligned}
 \text{Total stiffness} &= s_{\text{spring}} \times \text{No. of spring} \\
 &= 40 \times 8 = 320 \text{ N/mm}
 \end{aligned}$$

Axial force required to engage the spring;

$$W = 0.11 r_2^2 = 3970 \text{ Nmm}$$

$$\text{Initial compression in spring} = \frac{3970}{320} = 12.4 \text{ mm}$$

3. A single dry plate clutch is to be designed to transmit 7.5 kW at 900 rpm. Find the dia of shaft, mean radius, & face width of friction lining assuming mean radius to be the face width as 4, outer & inner radii of clutch plate, the face width as 4, assuming that no. of springs are 6 & dimensions of spring assuming that no. of springs are 6 & spring index = 6, the allowable shear stress for spring wire may be taken as ~~40 MPa~~ 40 MPa.

Ans. $T = \frac{\pi}{16} \tau_s D_s^3$
 $\tau_s = 40 \text{ MPa}$
 $P = \frac{2\pi NT}{60}$
 $7.5 \times 10^3 = \frac{2\pi \times 900 \times T}{60}$

$P = 7.5 \text{ kW}, N = 900 \text{ rpm}$
 face width = 4,
 no. of spring, $n = 6$.
 spring index = 6.

$\frac{7.5 \times 10^3 \times 60}{2\pi \times 900} = T = 22.795 \cdot 7 \approx 796 \text{ Nm}$

Design of belt drive.

Velocity ratio

It is the ratio b/w velocities of driver & follower.

Let d_1 be the dia of driver pulley, & d_2 be the dia of follower pulley, v_1 be the speed of driver in rpm, v_2 be the speed of follower in rpm.

→ Length of belt passing over the driver in one minute.

$$= \pi D_1 N_1$$

→ Length of belt passing over the follower in 1 min = $\pi D_2 N_2$.

$$\pi D_1 N_1 = \pi D_2 N_2$$

$$\frac{D_1}{D_2} = \frac{N_2}{N_1}$$

$$\frac{N_2}{N_1} = \frac{D_1 + t}{D_2 + t}$$

For multiple pulley systems:

$$\frac{N_4}{N_1} = \frac{D_1 \times D_3}{D_2 \times D_4} = \frac{\text{Speed of last follower}}{\text{Speed of first driver.}}$$

→ Product of dia of drivers
Product of dia of followers.

Slip of belt.

when, Frictional grip b/w belt & pulley becomes insufficient, it may cause some forward motion of driver without carrying belt over it. This is called slip. It is expressed in %.

Let slip b/w driver & belt = s_1 %.

slip b/w belt & follower = s_2 %.

Velocity of belt passing over driver per sec.

$$V = \frac{\pi D_1 N_1}{60}$$

$$\Rightarrow V = \frac{\pi D_1 N_1}{60} \cdot \frac{100 - s_1}{100}$$

$$\therefore v = \frac{\pi D_1 N_1}{60} \left[1 - \frac{s_1}{100} \right]$$

Velocity of belt passing over the follower.

$$\Rightarrow v = v \cdot \frac{s_2}{100}$$

$$\Rightarrow \frac{\pi D_1 N_1}{60} \left[1 - \frac{s_1}{100} \right] = \frac{\pi D_2 N_2}{60} \left[1 - \frac{s_2}{100} \right]$$

$$\frac{\pi D_2 N_2}{60} = \frac{\pi D_1 N_1}{60} \left[1 - \frac{s_1}{100} \right] \left[1 - \frac{s_2}{100} \right]$$

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \left[1 - \frac{s_1}{100} \right] \left[1 - \frac{s_2}{100} \right]$$

$$= \frac{D_1}{D_2} \left[1 - \frac{s}{100} \right]$$

$$\frac{N_2}{N_1} = \frac{D_1 + t}{D_2 + t} \cdot \frac{100}{100} \left[1 - \frac{s}{100} \right]$$

When the belt passes from slack side to tight side, a certain portion of the belt extends & it contracts when it passes from tight to slack side. Due to this change of belt length, there is a relative motion b/w belt & pulley surface. This is called creep. The effect of creep reduces the speed of driven pulley.

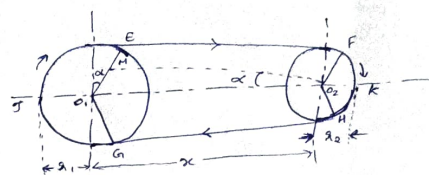
$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \cdot \frac{E + \sqrt{E_2}}{E + \sqrt{E_1}}$$

where, E = young's modulus.

σ_1 & σ_2 = stresses in tight & slack side.

Length of belt drive (open).

(Pg - 290)



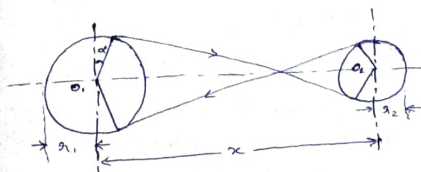
L = length of belt.

$$L = \text{Arc GE} + \text{Arc FH} + GH + EF$$

$$= 2 [\text{Arc GE} + EF + \text{Arc FH}]$$

$$L = \pi (r_1 + r_2) + 2x + \frac{(r_1 - r_2)^2}{x}$$

Length of cross belt drive.



$$L = \pi (r_1 + r_2) + 2x + \frac{(r_1 + r_2)^2}{x}$$

Power transmitted

$$P = (T_1 - T_2) v$$

w

$$\frac{T_1}{T_2} = e^{\mu \theta}$$

$\Rightarrow$ ratio of belt tension.

$$\text{centrifugal tension } T_c = m v^2$$

$$\text{Max tension, } T_{\max} = T_1 + T_c$$

Condition for max power transmission.

$$V = \sqrt{\frac{T_{max}}{3m}}$$

$$T_c = \frac{T_{max}}{3}$$

1. A flat belt is required to transmit 30kW from a pulley of 1.5m effective dia. running at 300 rpm. Angle of contact is spread over $11/24^{\text{th}}$ of the circumference. The coefficient of friction b/w belt & pulley surface is 0.3. Determine width of the belt required taking centrifugal tension into account. The given belt thickness is 9.5mm. Density of material is 1100 kg/m^3 & permissible working stress 2.5 MPa .

Ans. $P = 30 \text{ kW}$ $N = 300 \text{ rpm}$
 $d = 1.5 \text{ m}$ $\alpha = 360 \times \frac{11}{24} \times \frac{\pi}{180} = 2.88 \text{ rad.}$
 $\mu = 0.3$ $t = 9.5 \text{ mm} = 0.0095 \text{ m}$
 $\rho = 1100 \text{ kg/m}^3$ $\sigma = 2.5 \text{ MPa} = 2.5 \text{ N/mm}^2 = 2.5 \times 10^6 \text{ N/m}^2$

Velocity of belt, $V = \frac{\pi DN}{60}$
 $= \frac{\pi \times 1.5 \times 300}{60} = 23.56 \text{ m/s}$

$$\frac{T_1}{T_2} = e^{\mu \alpha} = e^{(0.3 \times 2.88)} = 2.37$$

$$P = (T_1 - T_2) V$$

$$30 = (T_1 - T_2) \times 23.5$$

$$T_1 - T_2 = \frac{30 \times 10^3}{23.5} = 1278.8 \text{ N}$$

$$T_1 - T_2 = 1273 \text{ N} \Rightarrow T_1 = 1273 + T_2$$

$$\frac{1273 + T_2}{T_2} = 2.375$$

$$1273 + T_2 = 2.375 T_2$$

$$1273 = 1.375 T_2$$

$$\therefore T_2 = 926 \text{ N}$$

$$T_1 = 2198.8 \approx 2199 \text{ N}$$

Mass of belt per meter length, $m = \text{Area} \times \text{length} \times \text{density}$.

$$m = b \times t \times \rho$$

$$= b \times 0.0095 \times 1100$$

$$m = 10.45 \text{ kg/m}$$

$$T_c = mV^2 = 10.45 \times 23.56^2 = 5805.8 \text{ N}$$

$$T = T_1 + T_c = \text{stress} \times \text{area}$$

$$= \sigma \times b \times t$$

$$= 2.5 \times 10^6 \times b \times 0.0095$$

$$5805.8 + 2199 = 2.5 \times 10^6 \times 0.0095 b$$

$$\therefore b = 125 \text{ mm}$$

2. Design a belt drive to transmit 10kW for a system consisting of 2 pulleys of dia 0.9m & 1.2m, centre distance of 3.6m & belt speed of 20m/s. $\mu = 0.3$, slip of 1.2% at each pulley & 5% frictional loss at each shaft & 20% overload.

Design of Gears.

Module-II

Spur gear terminologies.

- Pitch circle - Imaginary circle which pitch by rolling action would give the same motion as actual gear.
- Pitch circle diameter - Also called pitch diameter.
- Pitch point - Common point of contact b/w 2 pitch circle.
- Pitch surface - Surface of rolling disc which the meshing gear are replaced at pitch circle.
- Pressure angle - Also called angle of oblique. Angle b/w common normal of 2 gear teeth at the point of contact & the common tangent at the pitch point. (14.5° & 20°).
- Addendum - Radial distance of a tooth from the pitch circle to the top of the teeth.
- Dedendum - Radial distance of a tooth from root of teeth to pitch circle.
- Addendum circle - Circle drawn through top of the teeth. It is concentric with pitch circle.
- Dedendum circle - Circle drawn through the bottom of teeth & is concentric with pitch circle. Also called root circle. ($P.C.D \times \cos \phi$)
- Circular pitch - Distance measured on circumference of the pitch circle from a point of one tooth to the corresponding point on the next tooth.

$$P_c = \frac{\pi D}{T}$$

$D \Rightarrow P.C.D.$ $T \Rightarrow$ No. of teeth.

Conditions for meshing

If D_1 & D_2 are the dia of 2 meshing gear having

$$P_c = \frac{\pi D_1}{T_1} = \frac{\pi D_2}{T_2} \quad \text{or} \quad \frac{D_1}{T_1} = \frac{D_2}{T_2}$$

→ Diametral pitch - ^{ratio of} No. of teeth to the pitch circle dia in mm.

$$P_d = \frac{T}{D}$$

$$P_d \times P_c = \pi$$

$$P_d = \pi / P_c$$

$$P_d = \frac{1}{\text{Module}}$$

→ Module - ratio of P.C.D to no. of teeth.

$$\text{Module} = \frac{D}{T}$$

1, 1.5, 2, 3, 4, 5, 6, 8, 12, 16, 20, 25, 32, 40, 50 $\Rightarrow$ std. series of M.

→ Clearance =

Total depth

→ Path of contact

→ Length of path of contact

→ Area of contact

→ Arc of approach

→ Arc of recess

Imp. Q.

88-10-25.
1.

Rolling contact bearing.

Fraction of cycle.	Type of load	Radial (N)	Thrust (N)	speed (RPM)	service factor.
$\frac{1}{10}$	Heavy shocks.	2000	1200	400	3
$\frac{1}{10}$	Light shocks	1500	1000	500	1.5
$\frac{1}{5}$	Moderate shocks.	1000	1500	600	2
$\frac{1}{5}$	No shock	1200	2000	800	1

Select a single row deep groove ball bearing with operating cycle listed below, which will have a life of 15000 hrs. Assuming radial and axial load factors to be 1 & 1.5 respectively. An inner race rotates.

Ans. Given: $L_H = 15000$ hrs.

$W_{R1} = 2000$ N.	$W_{A1} = 1200$ N.	$N_1 = 400$ rpm.	$K_{S1} = 3$
$W_{R2} =$	$W_{A2} =$	$N_2 =$	$K_{S2} =$
$W_{R3} =$	$W_{A3} =$	$N_3 =$	$K_{S3} =$
$W_{R4} =$	$W_{A4} =$	$N_4 =$	$K_{S4} =$

$$X = 1, Y = 1.5, V = 1$$

Equivalent load, P for radial roller bearing,

$$P = X V F_A + Y F_R$$

(data book - 373) 16.7 B

$$W = (X V W_R + Y W_A) K_S$$

$$W_1 = (W_{R1} + 1.5 X W_{A1}) K_{S1} = 11000 \text{ N}$$

$$W_2 = (W_{R2} + 1.5 W_{A2}) K_{S2} = 4500 \text{ N}$$

$$W_3 = (W_{R3} + 1.5 W_{A3}) K_{S3} = 6500 \text{ N}$$

$$W_4 = (W_{R4} + 1.5 W_{A4}) K_{S4} = 4200 \text{ N}$$

Life of bearing in revolutions (L).

$$L = GONL_H = 60 \times N \times 15000 = \underline{0.9 \times 10^6 \text{ Revs}}$$

$$\left[\begin{array}{c} 3375 \\ 16.112 \end{array} \right]$$

$$L_1 = \frac{1}{10} \times 0.9 \times 10^6 \times N_1 = \frac{1}{10} \times 0.9 \times 10^6 \times 400 = \underline{36 \times 10^6 \text{ rev.}}$$

$$L_2 = \frac{1}{10} \times 0.9 \times 10^6 \times N_2 = \frac{1}{10} \times 0.9 \times 10^6 \times 500 = \underline{45 \times 10^6 \text{ rev.}}$$

$$L_3 = \frac{1}{10} \times 0.9 \times 10^6 \times 600 = \underline{108 \times 10^6 \text{ rev.}}$$

$$L_4 = \frac{1}{10} \times 0.9 \times 10^6 \times 800 = \underline{432 \times 10^6 \text{ rev.}}$$

$$\text{Equivalent dynamic load, } W = \left[\frac{L_1 W_1^3 + L_2 W_2^3 + L_3 W_3^3 + L_4 W_4^3}{L_1 + L_2 + L_3 + L_4} \right]^{1/3}$$

$$W = \underline{5767 \text{ N.}}$$

$$L = L_1 + L_2 + L_3 + L_4$$

$$\text{Dynamic load rating, } C = W \left(\frac{L}{10^6} \right)^{1/k} = 5767 \left[\frac{621 \times 10^6}{10^6} \right]^{1/3}$$

$$C = \underline{49.193 \text{ kN}}$$